Smart MCQ Solver Challenge

Milestone 4 Report

Fine-tuning BERT as a multiple-choice head with parameter-efficient LoRA adapters, then running inference.

DATASET	smart-mcq-solver-challenge — train.csv (2000 rows, 8 cols)
TASK	5-way multiple-choice question answering (options A–E)
MODEL	bert-base-uncased + LoRA (AutoModelForMultipleChoice)
SCOPE	10 checkpoint questions

1. Label encoding & input construction

Q1 Integer label at row 150

What it does. Maps letter answers A–E to class indices 0–4 and reads one encoded label.

How. Applied `LABEL_MAP` to the `answer` column; row 150 'C' → 2.

RESULT **`2 (answer 'C')`**

Q2 Character length of a prompt+[SEP]+option string

What it does. Builds one MC input by joining the prompt and option B with a [SEP] marker; measures its length.

How. `prompt + ' [SEP] ' + B` for row 0.

RESULT **`407 characters`**

2. Tensor shaping for multiple choice

Q3 Number of choices per question

What it does. Confirms each question is encoded as five parallel sequences (one per option).

How. Tokenized the five choice strings; reshaped to `(1, 5, 128)`; read dimension 1.

RESULT **`5`**

Q4 Total token positions in a 16-question batch

What it does. Counts every token slot when 16 questions are stacked.

How. Reshaped to `(16, 5, 128)`; `numel()` = 16 × 5 × 128.

RESULT **`10,240`**

3. Model head & loss

Q5 Logits produced per question

What it does. Checks that the multiple-choice head outputs one score per option.

How. `AutoModelForMultipleChoice` forward pass on `(1, 5, 128)`; logits dim 1.

RESULT **`5`**

Q6 Dimensionality of the loss tensor

What it does. Verifies the training loss is a single scalar.

How. Passed `labels` into the forward pass; loss (1.623) has `.dim() == 0`.

RESULT **`0 (scalar)`**

4. LoRA adapters

Q7 Trainable parameters under LoRA

What it does. Counts how few parameters are trained once LoRA adapters replace full fine-tuning.

How. `LoraConfig` (r=8, alpha=16) on query/value projections; summed `requires_grad` params.

RESULT **`295,681 trainable (0.27% of 109.8M)`**

5. Dataset assembly & training

Q8 Tokenized choices stored per item

What it does. Confirms each dataset record keeps its five encoded choices intact.

How. Built a dataset from the first 100 rows; first item's `input_ids` shape is (5, 128).

RESULT **`5`**

Q9 Final training step count

What it does. Runs a short LoRA fine-tune and reports where training stopped.

How. `Trainer` with `max_steps=4` over 32 rows (batch 4, seq len 64).

RESULT **`global_step = 4`**

6. Inference

Q10 Predicted probability for option E (row 0)

What it does. Runs the fine-tuned model on row 0; reads the softmax probability for option E.

How. Forward pass on the len-64 encoding; softmax over 5 logits, index 4. Probs are near-uniform (~0.20) after only 4 steps.

RESULT **`0.1924`**

Answer summary

Q	Checkpoint	Answer
1	Integer label at index 150	**2**
2	Char length (row 0, prompt+[SEP]+B)	**407**
3	Number of choices (dim 1)	**5**
4	Total token positions (16-batch)	**10,240**
5	Logits per question	**5**
6	Loss tensor dimensions	**0**
7	LoRA trainable parameters	**295,681**
8	Tokenized choices in input_ids	**5**

Q	Checkpoint	Answer
9	Final global_step	**4**
10	Probability of option E (row 0)	**0.1924**

Takeaway. Milestone 4 converts BERT into a trainable multiple-choice model and fine-tunes it with LoRA, updating only 0.27% of the weights. The near-uniform inference probabilities reflect the deliberately tiny 4-step demo run rather than a converged model.